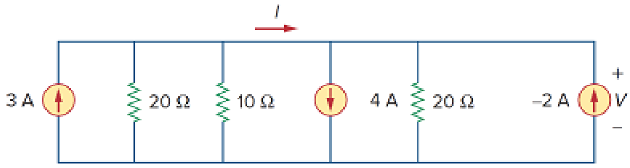
# Chapter 2, Problem 18P

## Problem

Find *I* and *V* in the circuit of Fig. 2.82.

**Figure 2.82:**




## Step-by-step solution

**Step 1** of 2

Refer to Figure 2.82 in the textbook.

From the circuit it is clear that all the elements in the circuit are connected in parallel between two nodes. Assume that the top node as "*V*" and bottom node as reference.

Apply Kirchhoff's current law at the node *V*.

$$-3+\frac{V}{20}+\frac{V}{10}+4+\frac{V}{20}+2=0$$
$$\frac{V}{20}+\frac{V}{10}+\frac{V}{20}=-3$$
$$V\left(\frac{1}{20}+\frac{1}{10}+\frac{1}{20}\right)=-3$$
$$V\left(\frac{1}{5}\right)=-3$$
$$V=-15\text{ V}$$

Therefore, the unknown voltage in the circuit is $\boxed{-15\text{ V}}$.

**Step 2** of 2

Apply Kirchhoff's current law at the node at which current "*i*" is entering.

$$i=4+\frac{V}{20}+2$$

Substitute $-15\text{ V}$ for $V$.

$$i=4-\frac{15}{20}+2$$
$$=4-0.75+2$$
$$=5.25\text{ A}$$

Therefore, the unknown current the circuit is $\boxed{5.25\text{ A}}$.